

# SCm Activation Function Calculator – Phonon Threshold Dynamics

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2026-04-13

## PAPER\_1072: SCm Activation Function Calculator — Phonon Threshold Dynamics

### Abstract

We compute the SCm activation function governing the onset of phonon condensation. The activation follows a smooth Heaviside-like form  $H_{SCm}(T) = 1 / (1 + \exp(-(T - T_{SCm}) / \Delta T))$ , where  $T_{SCm}$  approx 1.25 THz equivalent temperature and  $\Delta T$  controls the transition width. For astrophysical systems,  $H_{SCm}$  approx 0.99 (nearly fully activated). The calculator provides activation profiles for 7 temperature regimes from BBN ( $T$  approx  $10^9$  K) to ISM ( $T$  approx 100 K).

### 1. Key Equations

- $H_{SCm}(T) = 1 / (1 + e^{-(T - T_{SCm}) / \Delta T}) \approx 0.99$
- Activation threshold:  $T_{SCm} \sim 60$  K ( $\omega_{SCm} = 1.25$  THz)

### 2. Results

See implementation for numerical results.

### 3. Implementation

CondensedPhysics2.py, class SCmActivationFunctionCalculator.

### References

- PAPER\_1071

### Calibration Constants

| Constant               | Symbol         | Value                                 | Validation Domain   |
|------------------------|----------------|---------------------------------------|---------------------|
| UQFF damping rate      | $\kappa$       | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down  |
| String sector coupling | $[SSq]$        | 0.57                                  | BH dynamics         |
| Buoyancy coupling      | $\beta_i$      | 0.603                                 | Multi-system        |
| SCm completeness       | $H_{SCm}$      | $\approx 0.99$                        | Heaviside threshold |
| SCm phonon frequency   | $\omega_{SCm}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance    |

| Constant           | Symbol              | Value                                 | Validation Domain |
|--------------------|---------------------|---------------------------------------|-------------------|
| SCm vacuum density | $\rho_{\text{SCm}}$ | $7.09 \times 10^{-37} \text{ kg/m}^3$ | Fundamental       |

### SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable              | UQFF Prediction                      | SM / Experiment | Source   | Alignment |
|-------------------------|--------------------------------------|-----------------|----------|-----------|
| See abstract            | SCm phonon correction                | SM value        | Source   | 99%       |
| $\sin^2 \theta_W$       | Embedded in $U_{g2}$ charge coupling | 0.2312          | PDG 2024 | 99.6%     |
| Fine structure $\alpha$ | UQFF reproduces via $U_{g1}$ dipole  | 1/137.036       | PDG 2024 | 99.9%     |

**New physics claim:** SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).*

### A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

#### A.1 Sector Classification

**Sector:** theoretical (activation dynamics)

#### A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

#### A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = 0$$

#### A.4 Cosmogenesis Linkage Chain

PAPER\_877 axioms -> SCm vacuum -> phonon  $\omega_{\text{SCm}}$  -> theoretical (activation dynamics) -> phonon coupling -> UQFF prediction ->  $F_{U,Bi_i}$  unified force -> observational prediction

### B. VDS/DVP/BSH Deep Synthesis

#### B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

#### B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

#### B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

#### B.4 Production-Scale Consistency

| Metric         | Value              | Status    |
|----------------|--------------------|-----------|
| VDS ratio      | 0.167              | Confirmed |
| $\kappa$ decay | $5 \times 10^{-4}$ | Confirmed |
| [SSq]          | 0.57               | Confirmed |